

Bonus Task 1: camera orientation + alternate build

Camera orientation

Mounting the stereo/depth pair dead level is probably impractical to save Odysseus. Tilt it 15-20 degrees downward from the horizon: this splits the view between scanning the water surface (for Odysseus below the flight path) and the open horizon further out (for spotting something before the drone is close to him). If possible adding two narrower field of view passes: one forward level sweep during flight to the search area, one downward-tilted sweep during the actual grid search pattern beats a single fixed and sometimes compromised angle. This should be taken as a 2 phase mission profile as one would include a forward sweep and the other would include a downward sweep.

Alternate Build

Build 1 (Stereo IMX219-83 + Jetson Orin Nano) is my second choice as it finished a close second in the weighted score and is the build to fall back on if we decide the setection range and raw AI throughput matter more than the power/weight savings (if the search altitude needs to be higher, or the sea state means you need more robust detection against wave clutter).

Build 2 (the winner) optimizes more on endurance and simplicity.